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Recent vibration-based structural health monitoring on steel bridges: Systematic literature review

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ABSTRACT

Structural Health Monitoring (SHM) implementations on infrastructures, including bridges, are currently experiencing a surge in popularity as the favoured solution for preserving structural health. Additionally, the availability of a diverse array of SHM systems is a direct response to the increasing range of structures, the necessity to target specific damage mechanisms, the various monitoring objectives, the utilization of different sensor technologies, and the practical aspects of implementation. Hence, the research domain is challenged with the excessive state of subareas and integrations of techniques used which induced the need to organize the known knowledge about SHM practicalities, focusing on steel superstructure bridges and bridges with steel as main component. In order to gain a comprehensive understanding of this field and provide researchers with a clear perspective on SHM, it is essential to familiarize ourselves with the techniques that have been employed. To that aim, a thorough search was undertaken to locate publications dealing with terms such as "structural health monitoring," "vibration-based SHM," and "steel bridge/bridges". Scopus databases were checked for articles on vibration-based structural health monitoring for steel bridges and PRISMA procedure was conducted to apply systematic review process. The result highlights the profile of the publication trends under the topic throughout the year and the research phenomenon based on country of origin. Then, the result also reviews the studied steel bridge type to explore the propensity of vibration-based practicality based on structural configuration. Then, the heart of this paper, which is the third part reviews the methods that has been investigated by past researcher in the work of bridge vibration characterisation for SHM. The future of vibration-based methods is potentially tremendous as structure characterisation solution as it provides the state-of-the-art data-driven measures towards damage detection. However, the raise of SHM robustness has simultaneously raised the room for advancement towards achieving precise quantification and interpretation from vibration-based techniques. The room for advancements raised in this study has proven our one step forward to provide future studies of ways it can expand the vibration-based frontiers.

1. Introduction

Bridges are among a nation's precious assets for economical operation and services. Knowledge on bridge engineering and behaviour is expanding with vast establishments of methods not only to build but also to preserve bridge's safety. Even so, bridge collapse is still a common phenomenon these days which had posed negative reputation to a nation's transportation system and cause major disruption in economic growth [32,39,109]. Catastrophic failures also harmed public safety and cause worst case scenarios. For instance, the collapse of Silver Bridge took away 46 innocent lives in 1976 [51,69] while the failure of Quebec Bridge killed 75 people during its construction [93]. 43 deaths and 9 injuries were caused by an in-service collapse of Italian Morandi's Polcevera Viaduct with direct estimated economic losses of 100 million yuan [81,88]. Tuojiang Bridge's collapse has also caused another 64 deaths, 22 injuries along with estimated losses in terms of economic costing about 39.747 million yuan [94]. From these events of bridge failures, engineers and researchers became attentive to study the

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characteristics and causes of failures [37,128] and risks of failures related to bridge's substructure (see [122]). A bridge failures statistics study by [128] found that steel bridges comprise of the major proportion of failed bridges, among other material types. It is also well known that steel bridges are one of the earliest built type of bridges around the globe which has been in presence since the early 1900 s [25]. Thus, it is hypothetical that the remains of these ancient structures have experienced serious deterioration. In order to further assess the condition, comprehensive assessment methods are currently in demand for an accurate damage quantification [96]. In research domain, bridge health studies have penetrated the academia realm with introduction to many powerful solutions for structural health affairs, including Structural Health Monitoring (SHM) techniques [44].

The wave of Structural Health Monitoring (SHM) implementations on infrastructures including bridges as the solution for structural health preservation is at hype. Structural health monitoring (SHM) refers to the process of continuously or periodically assessing the condition and performance of a structure, such as buildings, bridges, dams, or aircraft, to ensure its structural integrity [3,15,20,130]. It involves the use of various sensors, data acquisition systems, and analysis techniques to monitor and evaluate the health of a structure over its lifespan. The importance of structural health monitoring lies in its ability to provide real-time or near real-time information about the structural behavior, performance, and potential damage of a structure [27]. Such robust and powerful integration of systems has aided the civil engineering industry in targeting multiple structural safety goals. For instance, bridge maintenance can shift from a reactive approach to a proactive one. By continuously monitoring the bridge, it is possible to identify and address potential issues before they escalate into major problems. This proactive maintenance approach improves the lifespan of the bridge, reduces downtime, and extends the time between major repairs or replacements. As addition, SHM also facilitates a better understanding of the bridge structural behavior and performance under different operational and environmental conditions [23]. By continuously monitoring the bridge, engineers can gather data on loads, stresses, vibrations, and other parameters that characterize the condition of the bridge. This data can be analyzed to optimize the design, assess the effectiveness for modifications, and ensure that the bridge is operating within its safe and optimal performance range. Currently, an SHM type called vibration-based SHM has proved its ability to tackle damage experienced by steel bridges [33,102]. Steel bridges, while durable and widely used in infrastructure, can experience various structural issues over time such as corrosion, fatigue, deterioration, and distortion. Vibration-based mechanisms made use of the mass properties input to define the modal parameters in two manners which are natural frequencies and mode shapes. Natural frequencies are proportional to a structural stiffness which can be affected if serious corrosion in steel structure takes place. When this happens, the monitoring of steel bridges shall alert on the shifts of the frequency's values. Vibration-based SHM can also aid in localizing the damage within the bridge. By analyzing the vibration data from multiple sensors, engineers can identify the regions or components of the bridge that exhibit anomalous behavior. This information helps focus subsequent inspections or targeted investigations to determine the exact nature and extent of the damage. In [72], the feasibility of a natural frequencies based SHM was investigated on a steel bowstring railway bridge in Belgium. During the monitoring period, the railway bridge has been retrofitted, resulting in data for two distinct states of the structure. A natural frequencies shifts was observed between the two states, quantifying the frequencies sensitivity on the bridge modifications. As addition, [57] also successfully demonstrated that a dual-criteria method based on a steel arch bridge modal parameter can detect and locate damage in the arch rib and vertical columns of deck-type arch bridges with considerable accuracy under a range of damage scenarios using only a few of the early modes of vibration.

SHM system exists as a wide domain of solutions which can be dissected through its nature of characterisation [106]. Some of the prominent SHM systems include vibration-based SHM, strain-based SHM, AI-based SHM, remote sensing and imaging, ultrasonic sensing and many others. Vibration-based SHM focuses on monitoring the structure using its vibration parameters. It typically involves the use of accelerometers or displacement sensors to measure the structural response to dynamic loads. Vibration-based monitoring is beneficial in providing valuable information about the structural behaviour using their modal parameters such as natural frequencies, mode shapes, and damping characteristics, which can be indicative of structural damage or changes in condition. SHM systems are also famously integrated with strain based SHM in order to advance monitoring goals to a multiple-dimension assessment [79]. Strain-based monitoring systems measure the strain or deformation in various structural components to assess their condition. Strain gauges or fibre optic sensors are commonly used to capture the strain distribution in critical areas of the structure. By monitoring changes in strain patterns, engineers can detect the development of cracks, excessive deformations, or material degradation, providing insights into the health of the structure [67,30]. Moreover, remote sensing and imaging SHM techniques involve the use of remote sensors or radar systems to assess the structural health of large-scale structures or infrastructure networks [54]. An in-depth review of bridge health monitoring focusing on the use of satellite-based multi-temporal synthetic aperture radar interferometry (MTInSAR) has been contributed [85], complemented with the application of MTInSAR for the portfolio-scale monitoring of bridges. Another novel succession of SHM using AI-based techniques for risk assessment of bridge heritage has also been demonstrated by [26] which extended the application of AI-based SHM on an existing RC bridge.

Overall, the wide range of SHM systems available is a response to the diverse nature of structures, the need to address specific damage mechanisms, the varied monitoring objectives, the availability of different sensor technologies, and the practical considerations of implementation. This variety ensures that there are tailored solutions for monitoring different structures under specific conditions, providing a comprehensive approach to structural health monitoring [99]. However, previous studies related to SHM suffered a gap in clearly setting the research topics direction. Numerous publications have been published in the past which discusses SHM methods for its novel successes as the solution to modern days structural preservation methods. Thus, there is a need to organize these articles to clearly obtain the way forward of each method. Researchers can also benefit through this article organization by finding study gaps in the method and application focus, only if a clear way forward or trends are delivered. Hence, this review paper is aspired to provide the insights on the recent development of vibration-based methods of SHM, focusing to solve structural problems faced by steel bridges. The author is confident that such comprehensive catalogue can assist interested researchers to gain understanding on methods available under vibration-based SHM, their condition of preferences throughout the years as well as their recent achievements in the field. The paper begins with introducing the method used to conduct the systematic review by using the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) method. Through the systematic review procedures on the articles, this paper will profile the selected articles in terms of their progression over the years and the research origin countries to visualize the studies done through these aspects. This paper also delves into the type of steel bridges which has evidenced the success of vibration-based methods for structural health investigations. Then, the next section, which is the heart of this paper, describes an analytic cataloguing on the explored vibration-based methods and their mechanisms on detecting bridge damage. Finally, the paper extends the purpose of the systematic review by finalizing the way forward of the methods based on the trends and the conclusions of the reviewed articles.

1.1. Research Gap, Objective, and research novelty

The growth in publications presenting bridge health studies has demanded the need to systematically compile the knowledge on this subject. Therefore, many SHM review articles has been published to serve that purpose and to help other researchers learn about the recent developments and accomplishments in the field [7,31,36,64] has systematically reviewed technologies implementation in construction industry in general without a focused technique. As addition, general reviews that focuses on specific approaches to detect damage has also been conducted, such as the application of vibration-based damage detection approaches in structures [41] and the successes of vibration-based methods in monitoring of civil infrastructure [21]. The author believes that these articles serve a good overview on vibration-based methods. However, a refined scope on specific type of structures is uniquely necessary for researchers focusing on damage detection of certain type of system. For example, it is an excellent work of [8] which had conducted a systematic literature review for SHM implementations focusing on girder-type bridges, with the aim to offer researchers a clear vision of this specific field by summarising techniques that have been used and the limits that exist in this line of research. In this paper, the author also highlights the nature of each method that has been preferred, which is the shortcoming of most results presented in previous similar studies. The SHM systems merged with Vibration-Based Damage Detection (VBDD) methodologies, statistical approaches, and signal processing techniques have also been reviewed by [97,117]. [9] then analysed the damage diagnosis systems for bridge structures with variety types of material in their review. A few review trends also focused on specific method under vibration-based methods. For example, Operational Modal Analysis (OMA) has been studied for its potential in SHM techniques and applications [75,59]. Referring to [126], OMA and ISMA's operational modal analysis approaches has been reviewed and described briefly through their progress over the years. From these records, vibration-based methods are seen to be gaining significant attention for review articles, but the scope was not focused to specific type of structure. By managing topic focuses, researchers can analyse trends and current interest favoured in a topic which somehow aids in finding research gaps. Systematic literature review is a solution that is able to serve such one-point guide for researchers who is considering venturing into a new topic of focus or for those who is interested in finding out the current trend and way forward of a research topic. Due to these benefits and contributions, the author is inspired to review the vibration-based methods specific on steel bridges application due to the material being one of the major options in bridge construction. Additionally, the scarcity of works on steel bridges condition assessment has also supported the need for systematic databasing of the topic. Moreover, SHM methodologies is one of the fast-moving subjects with continuous integration of systems from time to time. Thus, recent research works in structural health monitoring have breakthrough new methodologies in line with the invention of robust tools and sensors. Thus, a more updated review on vibration-based SHM is needed to update the domain with latest experimental mechanisms in order to comprehend new testing, analysis, and discoveries.

Understanding how SHM is systematised is the first step towards grasping how the topic is being coordinated in research. Therefore, it is beneficial to construct a taxonomy of the subject, especially if it is a wide topic. The publications are arranged according to a taxonomy of the works that have been published. Fig. 1 shows the literature taxonomy that helps to organise these diverse research fields into a meaningful, comprehensible, and cohesive framework. The construction of taxonomy can also help researchers find research gaps and study shortages in SHM-based bridge health monitoring. Researchers can gain valuable insights into the problem from the taxonomy's structure in several ways. In this way, the framework shows the subject's potential for further study and the potential space for adaptation in future research endeavours.

This article aims to present the implementations of vibration-based SHM on steel bridges from 2015 to 2022. Recent SHM methodologies for steel bridges were addressed, including research patterns by year and region, analysis used in past studies, bridge types assessed, and the future of vibration-based SHM deployment.

2. Research methods

This research analyses recent trends in SHM employing vibration-based approaches on steel bridge constructions. The approach of

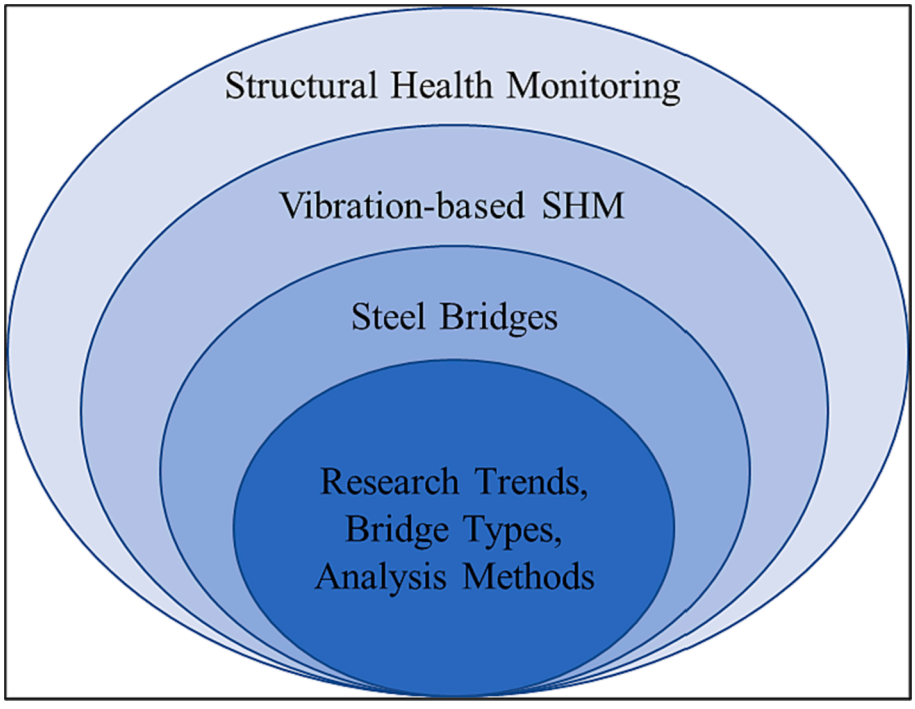


Fig. 1. Review systemisation of Vibration-based SHM on steel bridges.

conducting a systematic review of the relevant literature has been chosen to describe the developments in the research into the SHM phenomenon.

The available literature search followed the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) procedure guideline. The theoretical and methodological framework presented in the handbook is comprehensible and easy to pursue [80]. The approach was corroborated by [4]. The PRISMA was used for this review paper since the systematic dissemination following document screening simplifies the process for researchers to perform a thorough review. Moreover, PRISMA delivers more accurate estimations of findings than those produced from individual investigations [112]. This review will follow the PRISMA methodology, which requires three distinct phases: (1) accessing enormous databases of scientific, and academic literature through keyword and search techniques; (2) screening for inclusion and exclusion criteria; and (3) implementing an eligibility process for assessing relevant material to evaluate study data.

2.1. Selection criteria

The following list is the criteria used in the selection of publications: The inclusion criteria are as follows:

(1) This review only included English-language literature to assess the research's relevance.
(2) Only studies in structural health monitoring, vibration-based condition assessment, and vibration-based damage detection approaches were reviewed to better highlight vibration-based SHM especially applied to steel bridge constructions.
(3) The scope of these articles is SHM using vibrations on steel bridges.
(4) The article should include SHM trends, practical applications, and vibration-based SHM on steel bridge structures.
(5) Knowledge management papers not indexed by Scopus but referenced therein are also included.

The exclusion criteria are as follows:

(1) Structural health monitoring studies using non-vibration-based approaches are excluded.
(2) Not included are practical applications based on vibration that don't involve steel bridges. This covers SHM discussions about structures, including buildings, beams, slabs, and research that used laboratory models. No applications involving bridge structures are considered outside of the steel materials.

2.2. Paper selection phase

Step 1: Identification of Papers.

Scopus was used for the search due to its reputation for rigour and its inclusion in the selection criteria [24]. The Scopus Search engine was configured to conduct searching via title/abstract/keyword. The search criteria consisted of the phrases "structural health monitoring", "vibration-based", "steel" and "bridge" or "bridges." The first round of searching yielded 867 documents. Table 1 displays all of the specifics of the search parameters and the limiting mechanisms. All 867 scholarly papers were used in the publication retrieval process.

Table 1
Full search strategy and limitation functions used in Scopus.

Step	Criteria
1	"Structural health monitoring" and (vibration-based) and (steel) and ("bridge" or "bridges").
2	Limit to English language only.
3	The Year of publication is limited to 2015 until 2022 only.
4	Exclude all subject areas except engineering.
5	Document type is limited to "article" only.
6	The source type is limited to "journal" only.
7	The publication stage is limited to "final" only.

Step 2: Article Selection.

The articles were sorted for relevance using an article selection procedure. Abstracts were evaluated to determine which articles would advance to the next phase. Three criteria must be met for an article to be considered for inclusion in this review: 1) the article must explicitly relate to the implementation of SHM on steel bridges using vibration-based methods; 2) the article must focus at least on either a review of SHM trends, practical applications, or analysis linked to works of vibration-based SHM on steel bridge structures; and 3) featuring publications excluded from Scopus but cited in the selected literature. From the initial 867 retrieved academic publications, 67 passed the initial screening. Including all potentially relevant articles was ensured using a "snow-ball rolling" strategy; 11 additional publications were discovered by reading the references provided in the evaluated academic papers and then analysing the abstracts. Finally, the full texts of all 78 articles were used to conclude the following: research trends by year and place of origin; analyses adopted in previous studies; evaluated bridge types in previous studies; and the future of vibration-based approaches as efficient SHM implementation. A total of 789 articles were discarded due to this paper's exclusion criteria. Fig. 2 provides a flowchart of the PRISMA search phases.

Table 2 shows the percentage distribution of reviewed articles according to journal name. Relevant leading SHM journals such as Journal of Civil Structural Health Monitoring, Engineering Structures, Structure and Infrastructure Engineering, and Journal of Bridge Engineering were selected.

Step 3: Descriptive analysis and content analysis.

Following the identification of the papers, a comprehensive content analysis was carried out to (1) profile each paper according to the publication year and the origin nation; (2) characterise the research methodology and analysis employed; and (3) evaluate the path forward for vibration-based SHM on steel bridges.

3. Results

3.1. Profile of selected articles

3.1.1. Overview of SHM on steel bridges Prior to 2015

Despite steel bridges being one of the first types built on a global scale, status monitoring of steel bridges has received little attention since the early 2000 s. One of the early uses of vibration-based SHM on steel bridges was carried out by [6] for restoration and field testing of an FRP bridge deck on a truss bridge known as The Bentley Creek Bridge. Strain gauges were installed in the FRP deck and a steel floor beam to test the presence of composite action between the two and to measure the efficiency of the deck joints in transferring loads. Neither the deck nor the superstructure was found to exhibit any composite movement. Loads are transmitted via the longitudinal joint from one deck section to another, but the force is only partially transferred across the joint. The strain distribution of FRP decks may be affected by localised bending effects, according to test results. This research generated the notion of integrating strain data with vibrational information to pursue complex steel structure characterisation.

Owing to their cost-effective and straightforward installation, strain measurements are commonly taken with traditional strain gauges like the vibrating wire strain gauge and the semiconductor strain gauge. The design engineer can better use the derived stress or force values to assess the structure's state on the measured strain. Proficiency in remote and quantitative monitoring under authorised vehicles and fatigue investigations led [53] to build an in-service bridge monitoring system (ISBMS) to track live load stresses on a steel girder bridge in near real-time through the internet. One more novel idea in bridge monitoring employs strain values, as demonstrated by the work of [90], who

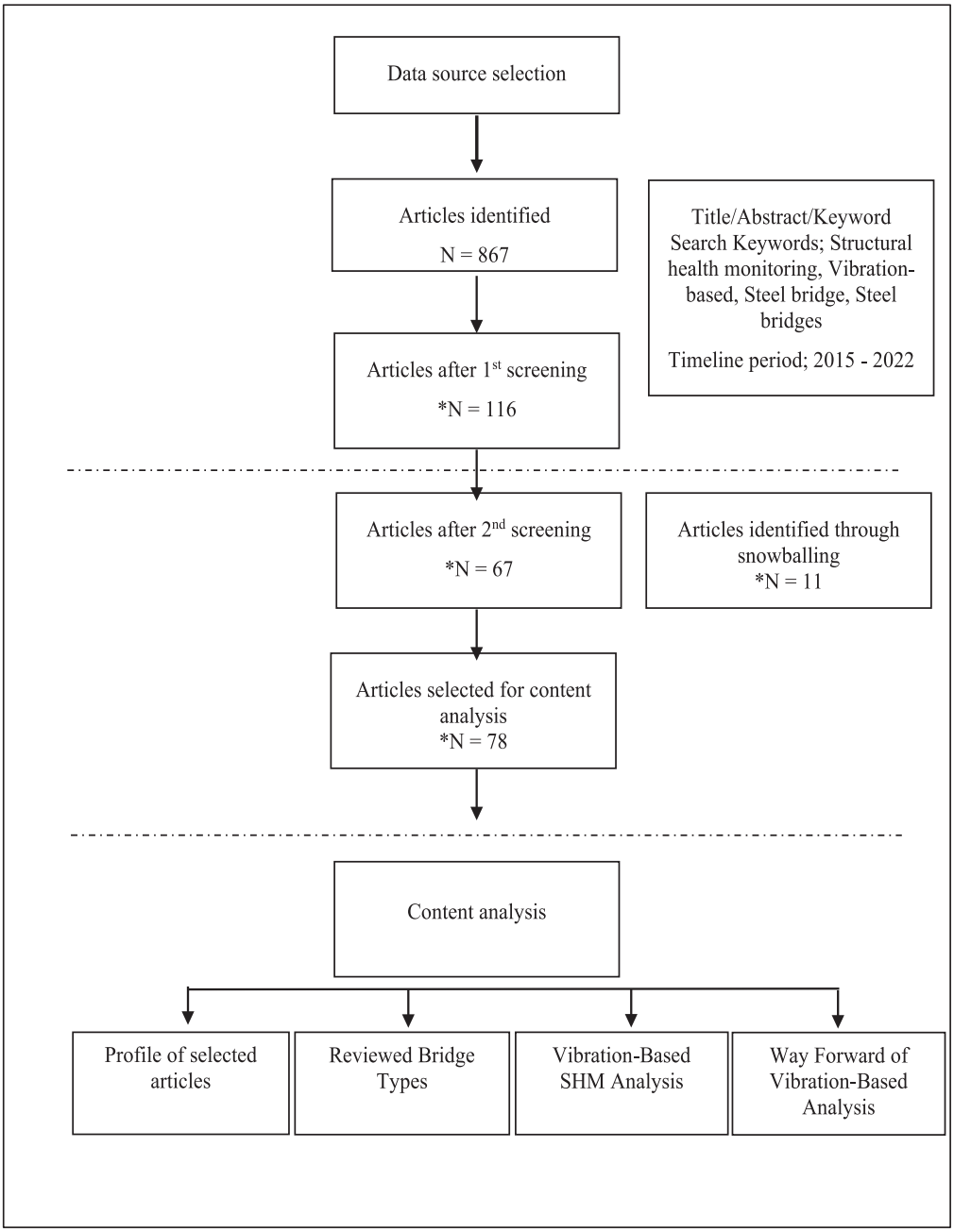


Fig. 2. PRISMA flowchart of the systematic review process.

incorporated a strain-based monitoring mechanism into the structural productivity evaluation of a steel girder bridge and subsequently used the data to carry out a structural reliability analysis.

Due to many sources of variability impacting recorded responses, damage features might alter even if no damage happens or remain unchanged when it occurs under actual world situations. This results in an inaccurate or non-existent detection of the damage. Since this ambiguity in testing and verification processes was not resolved, some publications expanded the vibration-based monitoring research to include other steel bridges [107]. For seismic evaluation and retrofit inquiry of the Henry Hudson Bridge, [48] analysed the data quality from ambient vibration testing of the bridge and noted a few potential flaws and uncertainties linked to ambient testing. The author(s) were able to verify data integrity with the help of comparisons to established standards and also discussed the range of allowable deviations.

Furthermore, [14] used full-scale static and dynamic testing to examine the riveted steel truss bridge's structural state and behaviour. Verification was performed by comparing measured values of maximum vertical deck displacement (static and dynamic) to theoretical values computed according to LST EN 1991–2/AC requirements [19]. The theoretical and experimental results are in excellent agreement, with an average discrepancy of only 2 %.

3.1.2. Trend analysis from 2015 to 2022

Few studies of vibration-based SHM have linked research findings to the context of steel bridges, as evidenced by the enormous gaps across reported publications before 2015. Since then, this tendency has made it difficult for researchers intrigued by the subject to locate relevant literature. Scopus publication analyser was applied to facilitate and highlight the investigation of this topic by year to statistically validate the observation. Publications were tallied using terms like "structural health monitoring," "vibration-based," and "steel bridge" to produce the

Table 2
Percentage distribution of reviewed articles according to journal name.

Journal name	Percentage, %
Journal of Civil Structural Health Monitoring	13.43
Engineering Structures	7.46
Structure and Infrastructure Engineering	7.46
Journal of Bridge Engineering	7.46
Structural Monitoring and Maintenance	5.97
Structural Health Monitoring	5.97
Mechanical Systems and Signal Processing	5.97
Sensors	5.97
Measurement: Journal of the International Measurement Confederation	4.48
Structural Control and Health Monitoring	2.99
Journal of Low Frequency Noise Vibration and Active Control	2.99
Structures	1.49
Smart Structures and Systems	1.49
Engineering Failure Analysis	1.49
Structural Concrete	1.49
Engineering Failure Analysis	1.49
Infrastructure Asset Management	1.49
Frontiers in Built Environment	1.49
Shock and Vibration	1.49
Baltic Journal of Road and Bridge Engineering	1.49
IEEE Transactions on Intelligent Transportation Systems	1.49
Journal of the South African Institution of Civil Engineering	1.49
Journal of Performance of Constructed Facilities	1.49
Case Studies in Engineering Failure Analysis	1.49
IES Journal Part A: Civil and Structural Engineering	1.49
Vibration	1.49
Dyna (Spain)	1.49
Acta Polytechnica	1.49
Journal of Sound and Vibration	1.49
Internet of Things (Netherlands)	1.49
International Journal of Structural Stability and Dynamics	1.49
Total	**100**

number indicated (Fig. 3).

According to Fig. 3, vibration-based SHM on steel buildings has advanced since it was first recognised in 2004. Nonetheless, there has been a significant increase in publications on the subject since 2015. Prior applications had focused on status assessment and damage detection by combining different methodologies to address issues in a wide range of structures, from individual structural components (like beams and plates) to entire architectural systems such as bridges and buildings [42]. After 2015, substantial analyses have been offered for additional innovative purposes, in contrast to the intent of practises before that year. The collected data undergo additional processes, such as verification and quality assertion. New developments in SHM methodology for steel bridges have necessitated substantial financial and institutional commitments to research. Fig. 4 displays the detailed annual documents released from 2015 to 2022.

Review articles covering the use of SHM for maintenance, management, and life-cycle assessment of structures remain a strong focus of the literature from 2015 through 2017. A total of 11 articles were assessed during this time, representing 15.5 % of all documents. Researchers tend to focus on a broader range of bridges, such as cable-stayed bridges and concrete bridges, for comprehensive characterisation of SHM technology, as is evidenced from these review papers [119,67,103,45,63,83,78,60,28,35].

However, several experimental methodologies based on structural vibration data have passed the test of time. Employing ambient vibration measurements on a steel truss bridge, [100] constructs a frequency response function using a linear, time-invariant transfer function T_{xy}, and conclude that it may be possible to distinguish and localise damages from the variations in the structure's modal characteristics. In addition, [108] also successfully developed a simple technique to pinpoint the damaged region of railway truss bridges depending on observed modal characteristics, addressing the insights of visual assessment, component testing, in situ load measurement, and finally, the construction of validated FEM. Besides, [105] performed a dynamic analysis of a 77-year-old single-span steel truss railway bridge utilising field measurement, modal analysis using a 3D finite element model, and a basic generalised

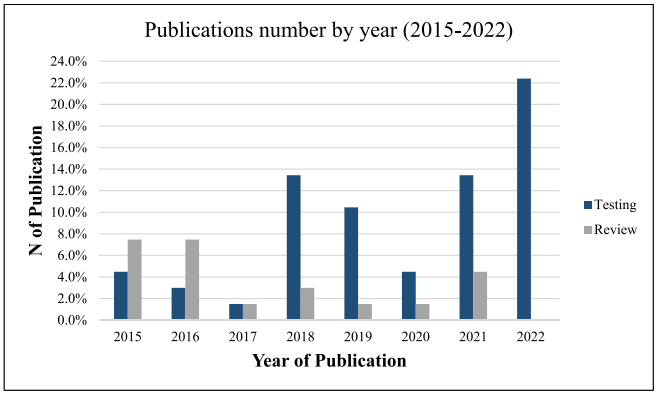


Fig. 4. The publications number plotted by year.

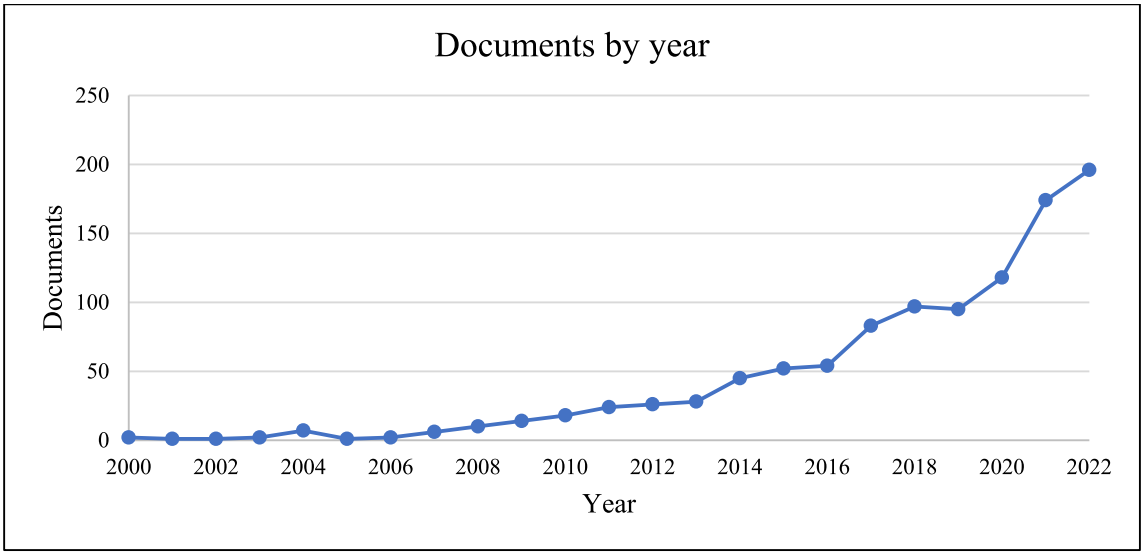


Fig. 3. The Scopus analyser result for topics involving structural health monitoring using vibration-based steel bridges.

single degree of freedom (SDOF) analysis. The bridge's natural frequency was accurately estimated by the SDOF model, finite element model with beam elements, and finite element model with shell elements. Furthermore, [29] used a stabilisation diagram-aided multivariate AR analysis of vehicle-excited bridge vibrations to investigate modal-parameter identification and vibration-based damage detection on a real-world, supported steel truss bridge with four simulated damage scenarios applied consecutively. The framework successfully detected the bridge's modal frequencies and mode shapes with outstanding precision, as seen by the minimal variance between independent test iterations and confirmed by a FE numerical model.

The year 2018 marks the beginning of the manifestation of forthcoming publications that contain more experimental investigations than review studies. In 2019, 7 experimental studies were applied to steel bridges, accounting for 10 % of all documents generated in Scopus, whereas in 2018, 9 experimental studies were credited, accounting for 13 % of all documents evaluated. The development of SHM has allowed for the cheap acquisition of acceleration measurements employed in estimating modal parameters, making SHM a valuable tool for damage localization at a low cost [108]. Studies conducted before 2018 have influenced a wide range of technical applications and in-situ testing of steel bridges as a result of this benefit, leading to a dramatic shift in SHM practises across the board, from the pre-testing phase to the post-processing phase.

SHM research is benefiting from the development of technology such as strain gauges, accelerometers, fibre optic sensors, inclinometers, and crack sensors [119]. However, the absence of a requirement for extensive cabling is an advantage of wireless sensor networks, which can help keep hardware costs down. These technologies are less stable and robust than wired systems due to radio signal propagation issues and battery replacement needs. Since then, researchers in the field of SHM have begun experimenting with wireless sensor networks to better support high-rate, real-time vibration detection [40,84,110,120]. Modern bridges are equipped with structural health monitoring systems that strive for long-term assessment rather than relying on traditional periodic testing [73,62]. Numerous algorithms have been developed to post-process data from SHM systems on steel bridges, including Bayesian network (BN)-based algorithm for recognising the pattern between modal frequency and multiple environmental elements [82], Kalman-type inverse approaches to predict the full-field response [95], progressive cross data analysis on dynamic responses [5], and particle swarm optimisation (PSO) and genetic algorithm (GA) [113]. The ancient bridge of Posadas (Córdoba, Spain) was erected in 1957, and [91] propose using a minimal number of sensors to monitor the bridge's structural health as an outlying target of historical appreciation research.

The rate of SHM installations on steel bridges is expected to increase rapidly over the next few years. Modal frequency changes in bridges induced by environmental and operational operations are a recent issue in the industry [131,89]. It was discovered that temperature and traffic loading are the key variables driving periodic changes in buildings. Removing the impacts of environmental and operational events on frequency variations is essential in developing vibration-based SHM to improve the precision of structural state assessment. Several efforts have also bolstered vibration-based methods for implicit damage identification. The Bayes factor, modal parameters, and their posterior uncertainties were successfully utilised to develop a novel Bayesian fast Fourier transform (FFT) approach for damage identification based on ambient vibration data [127].

A study by [57] introduced a dual-criterion method based on two damage indices that accurately detected and localised damage in an arch rib and vertical columns of deck-type arch bridges across various - damage situations with very few early modes of vibration. Damage in a suspension bridge's main cables and hangers was detected and localised with high accuracy by [121], who used the Modal Flexibility (MF) concept to develop and apply mode shape component-specific damage benchmarks such as the vertical damage index (DIV) and the lateral damage index (DIL). Another line of practical implementation research in the same year involved designing SHM systems on real bridges to offer valuable insights for design validation, structural model calibration, damage assessment, and operational decision-making assistance [104]. Derived based on the Hong Kong–Zhuhai–Macao Bridge, [123] carefully introduced the general purpose and structure of the SHM system, monitoring, and sensor information, which led to the development of a large-scale software to create a reliable and advanced monitoring system.

The global proclamation of rigorous proceedings and movement in numerous businesses in 2020 is hypothesised to be the impact of the COVID-19 epidemic when document production dropped significantly. Despite this, only a few pieces of research devoted explicitly to the topic have been satisfactorily reviewed. Many studies have used temperature impacts in health monitoring tools to further properly analyse structural conditions [43,40]. Research by [50] examined technical research about the health monitoring of civil structures under fluctuating temperatures. It provided an innovative process framework of damage assessment integrating temperature impacts into consideration to mitigate uncertainty issues brought on by climate conditions. Extensive research has been conducted on the involvement mechanism of uniform and non-uniform temperature on responses because of the proven association involving measuring responses and temperature from health tracking systems.

However, only some experimental research was undertaken during the same year. Research by [124] examined the present state-of-the-art objective performance and condition evaluation of a 30-year-old motorway bridge in New Jersey, integrating forced excitation testing, operational monitoring, visual inspection, non-destructive probes, controlled load testing, finite element modelling, long-term monitoring, and parameter identification within a structural identification framework. For the purpose of locating and approximating damages in steel girder bridges, [87] suggested a two-stage damage identification technique, the results of which indicated that all the damage spots were detected with adequate precision, even if several damage scenarios occurred in plate girders. Then, using data from multi-mode vibration recordings made under normal traffic conditions, [47] performed an additional SHM analysis on the dynamic properties of the highway cable-stayed Macau Rhine Bridge. It was discovered that the damping ratios can range from 30.1 % to 92.1 %, whereas the natural frequencies of the bridge modes can vary from 1.7 % to 5.5 %.

Few review papers were published in 2021 to extract a "generic paradigm" of Bridge Health Monitoring and aid newcomers, practitioners, and researchers through the different approaches proposed and developed to diagnose damage using data from sensors implanted in real structures, including steel bridges [98]. In the same year, wide range of experimental studies were also conducted which included bridge vibration analysis using FE model [13,18]. As addition, field vibration testing is another method that has been preferred lately in vibration-based methods to complement numerical computation results from FE bridge model. The recent preferences on this method may be based to the precision of the result, which depicts the actual or operational condition of the bridge at field.

For further analysis, [77] offered an unsupervised automatic data-driven methodology using time series analysis and multivariate statistical techniques to detect damage in railway bridges based on traffic-induced dynamic responses. Meanwhile, [16] proposed a unique method for vibration-based damage identification of civil structures and infrastructures under ambient stimulation. The work published by [61] used ambient and vehicle-induced vibration data from a steel truss bridge to aid in studying system identification and damage detection. In other recent publications, strain data has been investigated as a viable alternative to the more common acceleration-based monitoring for vibration analysis. This is because strain mode forms are more sensitive to changes in local stiffness than displacement mode shapes, and the

probability of a dense measuring grid may be obtained at a relatively low cost. Moreover, it was investigated by [10] how 80 Fiber-optic Bragg grating sensors may be installed on the deck of a steel-linked arch railway bridge. High-precision grating strain sensors are implemented by multiplexing sensors across four fibres and interrogating them using a unique method. In the same year, [71] provided data that included strain measurements were taken over 15 months on a steel bowstring railway bridge in Leuven, Belgium.

Stable SHM systems in the past years currently has been able benefit future studies by mechanising the established long-term bridge monitoring data after an ample collection duration [68]. A few studies in 2022 demonstrated improvisation and validation of SHM systems through simultaneity of methods with distinctive procedures. As instance, automated SHM has been worked on with various breakthrough frameworks as the glorification of SHM benefits towards the industry. [22] presented an attempt to establish a methodology for strengthening automated structural health monitoring procedures that relies on vibration spectrum analysis. The proposed approach involves the comparison of three methods of data processing: FFT, filtered FFT and AR. The experimentation of the methods on 6 bridges in Lithuania suggested that comparing to the traditional FFT and discrete FT analysis, the implementation of the AR procedure is much more promising in distinguishing the first resonant frequency from the obtained vibration spectrum. On the other hand, automated modal identification and tracking methods posed significant limitations for situation with very few sensors deployed and require manual intervention. [52] then proposed a unified fully automated modal identification and tracking method that can be used for structures with many or very few sensors deployed, for closely spaced modes with severe spatial aliasing and for structures with high flexibility or rigidity. In the study, a new index called Modified Modal Observability Correlation, was proposed to estimate the similarity of two modes. The proposed MMOC shows novelty in distinguishing different modes than MAC, and it can successfully distinguish different modes no matter whether the mode shape is wholly identified or not. It also can distinguish the closely spaced modes with severe spatial aliasing when very few sensors are deployed in a structure. The findings served ultimately novel furtherance of automated SHM as multiple mode elimination methods and automated modal identification and tracking methods are proposed to fulfil the fully automated modal identification and tracking in the area.

In the most recent year, modern tools has been actively configured into SHM to lengthen the functionality of technologic monitoring measures. We have seen the arrival of wireless sensor networks (WSNs) and wireless accelerometers technology in the past years, advancing in line with the increase of interest in vibration-based SHM. Developing and validating low-cost sensors is also critical for Structural Health Monitoring (SHM) of structures with limited funds for SHM assessments. Remarking the discussion, OMA has been successfully conducted using Low-Cost Wireless Arduino-Based Accelerometers in the work of operational and analytical Modal Analysis of a short-span bridge in Spain [65]. Battery-operated wireless sensors allow us to stop using wired sensor networks, providing easy installation processes and low maintenance costs. Nevertheless, wireless transmission of high-rate data such as structural vibration consumes considerable power. Consequently, these wireless networks demand frequent battery replacement, which is problematic for large structures with poor accessibility, such as long-span bridges. As an effort, [114] started experimenting a low-power multi-hop wireless sensor network suitable for monitoring large-sized civil infrastructures to handle this problem. The proposed network employs low-power wireless devices that act in the sub-GHz band, permitting long-distance data transmission and communication surpassing 1 km. The performance of the developed wireless sensor system was evaluated through a field vibration test on the landmark Manhattan Bridge in New York City which proved its reliability as an ideal configuration for monitoring large-size civil engineering structures and infrastructures.

Bridge collapses are still common in the modern world, even with improving condition monitoring techniques. In 2022, there are still a few bridge failures, including the Forbes Avenue Bridge (US), Loay-Clarin Bridge (Philippines), Tretten Bridge (Norway), Sanibel Causeway (US), and the deadliest case, the Julto Pul suspension bridge in Gujarat, India, which killed over 141 people. These mishaps demonstrate that SHM has not entirely replaced the field of structural assessment. Consequently, this paper provides context for the current SHM accomplishments and implementations, assisting interested industry players in understanding the significance of vibration-based SHM in pursuing damage detection objectives in steel bridges.

3.1.3. *Trend analysis according to country of origin*

Fig. 5 highlights that 22 different countries have made significant contributions to the experimental and review studies of SHM. The United States of America (USA) and China topped the list of countries that have made the most strides in structural condition assessment. Consider the four significant bridge collapses in the USA in 2015: the Pennsy Bridge, the I-10 Bridge, the Bob White Covered Bridge, and the Hopple Street Overpass collapse, which claimed the lives of one person. This may impact the abundance of literary research from the United States of America in the upcoming year of 2016. Besides, [83] reviewed the critical structural monitoring programmes recently established in the USA, focusing on large-scale structural system integrity and performance. In the same year, [63] examined recent research trends and efforts on SHM of civil infrastructure in Korea, concentrating on smart sensing, monitoring, and evaluation approaches of smart technology. Recent advances in integrating SHM with intelligent transportation systems (ITS) have shown the productive application of ITS devices (e.g., traffic cameras and detectors) in assessing bridge responses. Although [60] notes that SHM is not an impeccable system that accounts for considerable variability in diagnosis.

The ever-growing phenomenon of SHM studies in China, accounting for 20 % of all citations are idealized by the vast constructions of bridges

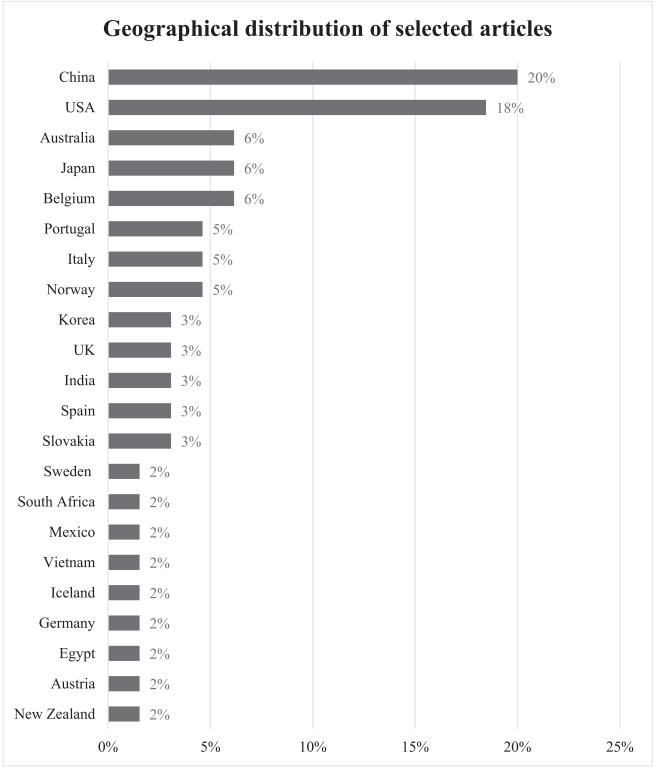


Fig. 5. Classification of Structural Health Monitoring on steel bridges papers that were published by origin countries between 2015 and 2022.

in their country. Solutions for addressing structural conditions on cable-stayed bridges have received extensive attention [127,67,73]. The Hong Kong-Zhuhai-Macao Bridge is a vast and complicated project. Moreover, [123] have introduced the overall purpose and structure of the SHM system, monitoring content, including sensor information cantered on this project. Large-scale SHM research has also focused on the Taizhou Bridge, with vibration-based wireless smart sensors mounted on the main cables of the massive three-pylon suspension bridge to determine the dynamic properties.

3.2. Reviewed bridge types

Many studies of bridge collapses take the form of statistical analysis or failure cause analysis [128]. The study also showed that steel bridges comprise the primary share of bridge breakdown when split by material kinds. For comparison, many of Malaysia's steel bridges in use today have exceeded their intended lifespan, they are in use in serving a variety of functions, including as a rail link. In the publishing years reviewed, Malaysia has not conducted SHM on steel bridges. Fig. 6 shows that the health monitoring tool may monitor many steel bridges, such as truss bridges, arch bridges, bowstring bridges, cable-stayed bridges, girder bridges, and suspension bridges.

Since 2015, research on truss bridges have obviously dominated structural health monitoring (SHM) [100,108]. Table 3 presented reviewed steel bridge types studied by previous researchers. From the 1870 s until the 1930 s, truss bridges were the most frequently constructed bridge style. There are still surviving examples of these bridges in various parts of the world, but the numbers are quickly dwindling as being replaced by more modern bridges. In the 1870 s, wrought iron bridges began to be constructed much faster in the USA than their wooden counterparts. The arched top chords of bowstring truss bridges were a popular truss design at the time. Concerns over the structure's health are undoubtedly on the rise in condition assessment episodes since its multiple truss bridge systems are in a condition that has prolonged its designed lifespan.

At the same time, cable-stayed bridges have received extensive attention, primarily from researchers from China, who have expressed considerable concern regarding this particular form of the bridge [73,85,118,123,129,131]. There has been a considerable increase in the number of cable-stayed bridges built in China during the past decade, illustrating this trend (i.e., the Sutong Bridge, the Stonecutters Bridge, and the Jiashao Bridge). Long-span bridges like cable-stayed bridges are preferred due to their rigid sturdiness, excellent aerodynamics, great seismic resistance, and aesthetically pleasing design. Despite their durability, cable-stayed bridges are vulnerable to environmental impacts, natural disasters, and long-term loads (like dead and vehicle loads). Bridge decks have been documented to have lost paint and deteriorated, piers and piles have corroded and scoured, and bearings,

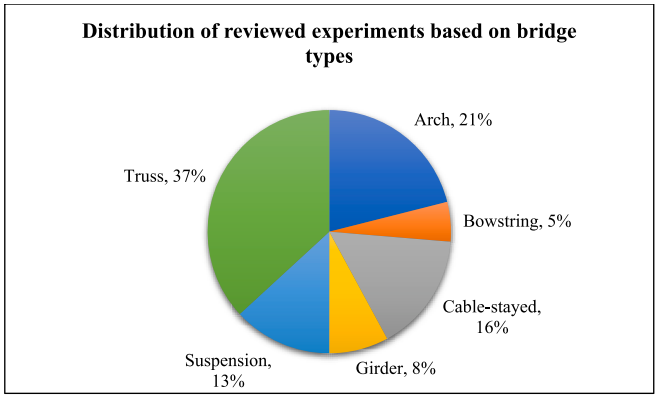


Fig. 6. The distribution of reviewed experiments based on bridge types.

Table 3
Reviewed steel bridge types studied by previous researchers.

Bridge Type	Authors
Truss	Saaed et al. (2015) [100], Siriwardane et al. (2015) [108], Shibeshi & Roth (2016) [105], Chang & Kim (2016) [29], Tran-Ngoc et al. (2018) [113], Lee et al. (2019) [66], Shahsavari et al. (2019) [104], Venglár & Sokol (2020) [115], Azim et al. (2021) [13], Kim et al. (2021) [61], AbdelRaheem et al. (2022) [2], Furtmüller et al. (2022) [46], Svendsen et al. (2022) [111], Venglár et al. (2022) [115]
Arch	Dammika et al. (2015) [35], Zhou et al. (2017) [132], Whelan et al. (2018) [120], Mu et al. (2020) [82], Jayasundara et al. (2019) [57], Anastasopoulos et al. (2021) [10], Anastasopoulos et al. (2022) [11], Bucinskas et al. (2022) [22]
Cable-stayed	Mao et al. (2018) [73], Alamdari et al. (2019) [5], Yan et al. (2019) [123], Zhou & Sun (2019) [131], Górski et al. (2020) [47], Ni et al. (2021) [86]
Girder	Kim et al. (2018) [62], O'Higgins et al. (2019) [89], Nick et al. (2021) [87]
Suspension	Petersen et al. (2019) [95], Wickramasinghe et al. (2020) [121], Li et al. (2022) [68], He et al. (2022) [52], Tronci et al. (2022) [114]
Bowstring	Maes & Lombaert (2021) [71], Maes et al. (2022) [72]

expansion joints, and other motion gear have malfunctioned [67].

3.3. Vibration-Based SHM analysis adopted by published literature

SHM and non-destructive inspection data provide definitive insight into the building's existing conditions [45]. It can serve as an early warning system for structural damage, allowing preventative measures to be taken before costly repairs are needed. Among SHM's many goals are identifying the real structural reactions under normal operational loads. Other goals include identifying degradation in multiple building elements, forecasting structural performance, and optimizing management interventions. The data gleaned from SHM and non-destructive testing is then invested wisely by being integrated into improved performance forecasting models.

Several SHM algorithms have been investigated for their capability to provide relevant context for damage assessment. Scores of analyses have bolstered the value of vibration-based SHM by facilitating the incorporation of structural dynamic information. Table 4 summarises the sorts of analyses carried out by existing literature in the process of vibration-based SHM.

a) Finite Element Method (FEM): Many research projects incorporate Finite Element Analysis for modal analysis and field data verification [108,34,105,132,91,56,113]. Several authors have also used FEM to create a comprehensive bridge baseline model, which visualises the bridge in its post-construction initial state [77]. The previously validated models can be technically expanded to investigate vibration-based damage detection (VBDD). An FE model was created by [121] and is regarded as the whole baseline model for VBDD implementation. The study tested the proposed damage identification process by adding simple damage events to the FE model. Additionally, [87] developed and verified an FE model predicated on the actual bridge's modal features. The approved FE model of the actual bridge was then used in a damage identification technique. On the other hand, [18] created 14 FE models of the same bridge, each with distinctive characteristics such as a reduced number of stay cables, a modified arrangement of the cables, a greater curved deck, and a higher inertial tower cross-section. The analysis represents a novel application of finite element analysis (FEA) to the study of vibration-based anatomy, intending to quantify the effect of design decisions on cable-stayed bridge's modal parameters.
b) Operational Modal Analysis (OMA): In recent decades, vibration-based operational modal analysis (OMA) employing ambient data from the SHM system has gained interest. OMA is a modal parameters identification approach used to determine a structure's modal parameters from vibration data acquired under operational

Table 4
Vibration-based research methods conducted by previous researchers.

Types of Analysis	Authors
Finite Element Analysis	Siriwardane (2015) [108], Dammika et al. (2015) [35], Shibeshi & Roth (2016) [105], Zhou et al. (2017) [132], Pachón et al. (2018) [91], Jamali et al. (2019) [56], Tran-Ngoc et al. (2018) [113], Whelan at al. (2018) [120], Venglár & Sokol (2020) [115], Jayasundara et al. (2019) [57], Shahsavari et al. (2019) [104], Wickramasinghe et al. (2020) [121], Nick et al. (2021) [87], Kim et al. (2021) [61], Briseghella et al. (2021) [18], Azim & Gül (2021) [13], Anastasopoulos et al. (2021) [10], Meixedo et al. (2021) [76], Anastasopoulos et al.(2022) [11], Furtmüller et al. (2022) [46], Svendsen et al. (2022) [111], Maes et al. (2022) [72], Venglár et al. (2022) [116]
OMA - SSI	Zhou et al. (2017) [132], Alamdari et al. (2019) [5], Tran-Ngoc et al. (2018) [113], Whelan et al. (2018) [120], Venglár & Sokol (2020) [115], Zhou & Sun (2019) [131], Jayasundara et al. (2019) [57], Wickramasinghe et al. (2020) [121], Maes & Lombaert (2021) [71], Anastasopoulos et al. (2021) [10], He et al. (2022) [52], Anastasopoulos et al. (2022) [11], Maes et al. (2022) [72], Tronci et al. (2022) [114], Komarizadehasl et al. (2022) [65], Venglár et al. (2022) [116]
OMA - FDD	Shahsavari et al. (2019) [104], Briseghella et al. (2021) [18], Li et al. (2022) [68], AbdelRaheem et al. (2022) [2], Svendsen et al. (2022) [111], Tronci et al. (2022) [114], Komarizadehasl et al. (2022) [65]
OMA - EFDD	Wickramasinghe et al. (2020) [121], Nick et al. (2021) [87], Meixedo et al. (2021) [76]
Model Updating	Pachón et al. (2018) [91], Tran-Ngoc et al. (2018) [113], Whelan at al. (2018) [120], Shahsavari et al. (2019) [104], Svendsen et al. (2022) [111]
Fast Fourier Transform (FFT)	Siriwardane (2015) [108], Zhang et al. (2021) [127], Yan et al. (2019) [123], Nick et al. (2021) [87], Ni et al. (2021) [86], Bucinskas et al. (2022) [22], Mao et al. (2022) [74]
Short-time Fourier transform (STFT)	Furtmüller et al. (2022) [46]
Discrete-time Fourier transform (DTFT)	Venglár et al. (2022) [116]
Frequency Response Functions (FRF)	Saaed et al. (2015) [100]
AR Analysis	Chang & Kim (2016) [29], Bucinskas et al. (2022) [22]
Random Decrement (RD) Technique	Zhou et al. (2017) [132], Yan et al. (2019) [123]
Eigensystem Realization Algorithm (ERA)	Dammika et al. (2015) [34]
Modal Damping Analysis	
Single-degree-of-freedom (SDOF)	Shibeshi et al. (2016) [105]
Hilbert Huang Transform (HHT)	Mao et al. (2018) [73], Yan et al. (2019) [123]
Data-driven Random Subspace	Mu et al. (2020) [82]
Kalman Filter (KF), Wavelet Transform (WT), S transform (ST), Multiple Signal Classification (MUSIC), Blind Source Separation (BSS).	Yan et al. (2019) [123]

conditions. Other names for OMA include ambient modal identification, output-only modal analysis, and in-operation modal analysis. A variety of data processing algorithms form the foundation of OMA methods. OMA data interpretation is complex and can easily result in misleading outcomes despite appearing to lead to reasonable ones [17]. In the last 30 years, scientists have developed various OMA techniques to tackle this issue.

i. Stochastic Subspace Identification (SSI): Compared to other OMA techniques, stochastic subspace identification (SSI) is favoured due to its high parameter estimate accuracy and great computing efficiency in the time domain. Few authors have used SSI to do operational modal analysis in previous research due to its mathematical complexity [5,120,57,115,131,10,71]. The SSI can be implemented in two ways: data-driven (SSI-data) and covariance-based (SSI-cov). As [38] noted, the SSI-cov is less computationally intensive and easier to pursue than the SSI-data. Even so, the SSI-cov implementation achieved the same level of accuracy as SSI-data. Consequently, SSI-cov was the tool to carry out the dynamic system identification of the examined bridge [113]. There have been studies that used SSI parameter identification alongside other techniques for the goal of making comparisons [132]. The vibration parameters were extracted by [121] using two distinct operational modal analysis (OMA) techniques: Enhanced Frequency Domain Decomposition (EFDD) and Stochastic Subspace Identification (SSI). The mode extraction procedure in SSI was based on the principle component (PC) algorithm. The modal assurance criteria (MAC) values were calculated to perform a correlation analysis between the two methods, and a total of nine modes were retrieved using the techniques.

ii. Frequency Domain Decomposition (FDD) – As with Peak Picking (PP) and Enhanced FDD, FDD belongs to the frequency domain OMA methods category. It is a popular OMA technique that detects mode multiplicity using singular value decomposition (SVD) of the power spectral density (PSD) matrix [126]. Over the past few years, there has been limited experimentation with adopting this method to apply OMA on steel bridges. Modal analysis utilising FDD has been successfully performed on Memorial Bridge (US) by [104] to calibrate the structural model for use in the structural performance assessment. Moreover, [18] took an innovative approach by using FDD to determine the dynamic response of a curved cable-stayed bridge in Venice (Italy).

iii. Enhanced Frequency Domain Decomposition (EFDD) – EFDD is a processing approach that is user-friendly and time-efficient. It provides a highly accurate approximation of the natural - frequency and the frequencies of closely spaced modes. However, [1] remark that accurate computation of modal damping remains a challenging unresolved problem, frequently resulting in inaccurate estimations. Research on a limited number of bridges indicates that EFDD yields excellent modal identification results [77]. In the study by [121], EFDD and SSI were used to extract the vibration information of the subject being examined. The results revealed a good correlation between the two approaches, except for one mode. EFDD could not recognise the missing mode despite linear interpolation's missing values. This paper compares frequencies and mode shapes retrieved from SSI to those derived from the FE model.

c) Model Updating – Many different strategies for model updating have been verified and documented [91,104]. Conventional methods often involve minimising some structural qualities to reduce some error functions between the observed and calculated responses. The success of these approaches relies on a few factors, including the availability of a meaningful error function, an appropriate numerical model, a precise determination of the experimental parameters, and the application of an efficient optimization algorithm to locate the global minimum of this function. Most studies have employed gradient-based methods to update finite element models of bridge structures. In contrast, only a handful has looked into using genetic algorithms for the structural identification of full-scale bridge constructions [120]. Meanwhile, [113] showed that a credible model for the bridge's health condition evaluation and operational safety

management required considerable vibration monitoring and model updating.

d) Fast Fourier Transform (FFT) – An easy-to-implement and practical, fast Bayesian FFT approach for identifying ambient modalities have recently gained popularity [123,87,86]. Numerous constructions, including skyscrapers, bridges, and more than 100-foot-tall buildings, have benefited from its use in modal identification. By measuring the modal parameters of railway truss bridges, for example, [108] applied FFT to obtain mode displacements to pinpoint the damaged area. Likewise, [127] have recently invested time and effort into constructing a novel Bayesian fast Fourier transform (FFT) strategy for damage detection utilising the Bayes factor on ambient vibration data. The FFT method was used to visually check the frequency and time domains to assess the quality of the collected data [132].

e) Frequency Response Function (FRF): The input force and response measurements are necessary to extrapolate the frequency response function (FRF) for use in the modal structure analysis [12,49]. It is usually tricky to estimate the input force for large civil engineering constructions since these structures are typically built using huge materials. Modal characteristics can be estimated with sufficient accuracy using either the output-only response or ambient vibration measurements. According to [100], the linear, time-invariant transfer function T_{xy} was efficaciously applied to construct FRF based on ambient vibration measurements to explore alterations in modal characteristics before and after railway bridge damage.

f) AR Analysis: Auto-regressive analysis allows for highly precise and accurate identification of the bridge's modal frequencies and mode shapes. Using a stabilisation diagram-aided multivariate autoregressive analysis, [29] extracted modal parameters from vibrations triggered by passing vehicles. Given that the modal parameters are used in place of the damage-sensitive features, the two processes are essentially interchangeable for extracting the former. Thereby, a multivariate autoregressive (AR) time-series technique supported by stabilisation diagrams was applied to achieve this task automatically by choosing relevant structural modes.

g) Random Decrement (RD) Technique: The random decrement technique (RDT), is applied to every mode of modal frequencies to determine the correlating damping ratio. The core of RDT is an averaging procedure, also known as a random decrement procedure. After combining the random decrement (RD) method with the complex mode indicator function (CMIF) to remove the impact of epistemic uncertainty, [132] compared the findings to those obtained using the SSI method. The work of designing and implementing an SHM system for a massive sea-crossing project involving bridges and tunnels also made use of RDT in association with Fourier transform (FFT), short-time FFT (SFFT), wavelet transform (WT), and another dynamic characterisation method.

h) Eigensystem Realization Algorithm (ERA): The eigensystem realisation algorithm is developed to identify modal parameters and reduce models of dynamic systems based on experimental data. The minimum order realisation method, an improved version of the Ho-Kalman algorithm, was introduced alongside the singular-value decomposition method. A study by [58] provides a comprehensive overview of ERA for modal identification. Besides, [34] showed how vital ERA is in calculating the steel arch bridge's natural frequencies, mode shapes, and modal damping ratios. The damping ratios and modal identification results vary for different system orders. This emphasises the significance of thoroughly extracting the modal damping ratio values for enhanced diagnostic precision. The modal amplitude coherence (MAC) [58] is a crucial quantity in ERA that represents the association between the modal time series reproduced by the discovered eigen variables and the modal element of the original time series, ergo measures the precision and consistency of the identification. Despite this, MAC is indifferent to imprecise identifications. As a result, the findings found by ERA were screened using the extended MAC (EMAC) [92].

i) Analytical Damping Evaluation: A study by [34] introduced a novel energy-based damping model to predict the damping parameters of a steel arch bridge, including comparable loss factors of structural elements, and then analytically evaluated the bridge's modal damping ratios. The proposed analytical method can supplement the diagnosis of damaged parts of steel bridges by experimental damping identification. The literature reveals that when local members of a steel truss bridge are damaged, the modal damping ratio of coupled modes of local members considerably increases [125]. The suggested energy-based technique allows for an analytical evaluation of experimentally detected damping changes, enhancing the capability of identifying the sources of damping change. Along with the increments in damping, it is possible to anticipate substantial changes in the damaged component's damping parameters. Thus, it was determined that the approach can quantify the damping contributions of various damping sources to the bridge's modal damping ratios.

j) Single-degree-of-freedom (SDOF): As one method for doing dynamic analysis on a 77-year-old single-span steel truss railway bridge, [105] opted for a simple generalised single degree of freedom (SDOF) analysis. The outcomes exhibited that the SDOF model provided reliable displacements when the effects of the axle were added to the model. Several models, including the FEM with beam elements and the FEM with shell elements, yielded relatively relevant predictions for computing the fundamental natural frequency of the bridge.

k) Hilbert Huang Transform (HHT): A few researchers have used the Hilbert-Huang Transform (HHT) to determine modal frequencies and have independently confirmed that HHT is an efficient modal identification approach due to its anti-noise and adaptive properties [73]. The requirement for a comprehensive description of non-linear distorted waves and the variations in these signals inherent to non-stationary processes drove the creation of HHT. First, [73] effectively evaluated HHT to calculate the one-year worth of continuous structural natural frequencies of the Sutong Cable-stayed Bridge in China. Then, using the HHT, [123] separated a signal recorded from the Hong Kong-Zhuhai-Macao Bridge into a finite set of intrinsic mode functions (IMFs). The study also compared HHT to Fourier transform (FFT), short-time FFT (SFFT), wavelet transform (WT), S transform (ST), fast ST (FST), Hilbert transform (HT), multiple signal classification (MUSIC), and blind source separation (BSS), declaring it relevant for its good time–frequency resolution, high signal–noise ratio, and ease of implementation.

3.4. Way forward of Vibration-Based analysis for successful structural health monitoring

Various types of vibration-based SHM experimentations has been reviewed. From the articles, noteworthy anatomies were recorded to enhance SHM significance as a powerful mechanism in structural engineering industries. Primarily, traditional techniques for structural health monitoring rely heavily on visual observation and non-destructive testing [108]. Visual inspections lack reliability since it is contingent on the skill and knowledge of the inspectors. Besides, non-destructive testing procedures are also deemed to be somewhat pricey. In vibration-based techniques, dynamic features, including natural frequencies, mode shapes, damping ratios, and other variables, are applied to assess the structure's condition [60]. In addition, structural vibrations have been used for various purposes, with most focus on vibration-based damage detection (VBDD) [70,103,121]. This approach can identify damage on a global scale by deploying sensors not particularly close to the unknown damage location. The amount of information offered by the recorded responses merged with robust analyses discussed in the preceding sections determines the level of precision possible in

damage identification. This section examines SHM's development of dynamic features as status assessment tools.

Furthermore, [5] introduced a novel incremental multi-way data processing algorithm to analyse the dynamic response of a cable-stayed bridge in Western Sydney. This is one of the most notable demonstrations of vibration-based techniques for efficient condition monitoring. When applied to OMA with SSI-cov, the method achieved a remarkable F1 score of 99.75 % (high accuracy) in detecting simulated bridge damage. It could also measure damage severity, locating it in most of the examined scenarios. Another paper offers a thorough evaluation of damage detection methods that account for changes in natural frequencies [55,101,40] also published a case study on using autonomous Wireless Smart Sensors (WSS) to monitor the major cables for vibration on a massive three-pylon suspension bridge in China. The study also yielded outcomes for an anti-aliasing filter, a statistical analysis technique, and a technique for estimating cable stress as potential solutions. Regarding vibration monitoring of the main cable, WSSs are adaptable and self-sufficient, with only the results needing to be transmitted to the gateway nodes. The range of the vibration magnitude of the main cables was found to have increased slightly over 19 months, according to the analysis of the main cable vibration data. This is primarily due to the growing number of vehicles crossing the bridge over time and the implications of the constrained long-term reliability of the accelerometers. These results corroborate the work of [103] that the parameters measured in a vibrational manner are strongly correlated with early damage occurrences. Changes in the dynamic qualities of bridges are also heavily influenced by external elements, including weather, traffic, and wind. The dynamic features of the Sutong Cable Stayed Bridge (SCB) were examined by employing one year of continuous monitoring data under working settings by the SHM system. This data included acceleration and strain responses and modal frequencies [73]. The HHT approach is then used to determine the girder's modal frequencies at various times of the year in the vertical and torsional directions. The study found that, among environmental factors, the temperature is the most influential on the modal frequencies of SCB, particularly for the higher-order modes.

According to [100], changes in the modal properties of the structure generated from measured vibrations caused by train traffic can be used to detect and pinpoint defects in steel railway bridges. The presence of damage was verified by comparing the bridge's healthy and collapsed states across a range of frequencies, damping ratios, and mode form. The data also revealed substantial shifts in the bridge's modal properties as deterioration progressed. Modal damping is a measure of how much a structure can absorb or transmit vibrations, and an abnormal proportion of change between the healthy and any other state for a structure can be regarded as a serious signal for the early stages of deterioration. An indicator of the presence of severe damage in steel bridges is a 206 % increase in the damping ratio comparing a healthy bridge to a collapsed one. Research by [29] applied four artificial damage scenarios in sequence to a real, essentially supported steel truss bridge. Modal frequencies and mode shapes of the bridge were found with great precision and accuracy using the stabilisation diagram-aided multivariate AR analysis of vehicle-induced bridge-free vibrations. The accuracy was confirmed with a FE numerical model, and the small variation between test runs highlighted the precision. Modal frequencies decreased as damage causing a high-stress redistribution was adhered, denoting a loss of global stiffness. When the damage was applied asymmetrically, it altered the mode's form in symmetric and anti-symmetric ways. Although these findings are limited to bridge constructions and damage situations identical to those in this experimental investigation, the methodologies herein suggest the prospective genuine application of deployment.

As conclusion, vibration-based SHM technique has been established to be a valuable tool for assessing a structure's durability, thus reasoning to the specific focus towards upgrading the system tools to enable low-cost operation in the recent years. However, further research is recommended to enhance SHM by integrating vibration-based components and analysis, empowering engineers with greater expertise in this domain. From the review, a few shortcomings rise to induce gaps that can be argumentative for future works. First, it is observed that some studies may jeopardize the accuracy and reliability of the analysis from the lack of long-term data on tested structures. Depending on the scale of the investigation, this condition may be related to the cognition that implementing SHM systems can be costly, especially studies including retrofitting existing structures. SHM also relies on the precision of result interpretation which evolves around the need to establish accurate baseline data. However, obtaining and maintaining reliable baseline data for comparison can be challenging, especially for older structures or those with unknown historical data. The solidity of SHM studies found their pathway in data-driven domains which it enables quantification of discrepancies. Thus, the baseline module is critical towards resolution because it facilitates the early detection of damage, which is also the strength of SHM. Moreover, the literatures assessed in this study also yielded that the analysis of large amounts of vibration data can be computationally intensive, requiring powerful computing resources and skilled personnel. The computational complexity has advanced linearly with the robust state of the solution provided. As such, the acknowledgement of these shortcomings is already a come to fore of the topic's evolution. Researchers in the area are definitive that firstly, joint ventures between academia, industry, and government agencies are potent factors to pool resources, knowledge, and expertise for more effective implementations in the area. Simultaneously, the ventures are only coherent to SHM explorations because it elevates the chance to increase the availability of comprehensive and long-term data by deploying a network of sensors strategically placed on the structure to capture a wide range of data, which induce precise baseline reference. Finally, there is also a light in stabilizing the spark of SHM preferences through standardization and benchmarking of the method. Establishment of industry standards and benchmarks for SHM methodologies, ensuring consistency and comparability across different studies may be the solution to capture the composite nature of SHM mechanisms. Besides, establishing industry standards and benchmarks for SHM methodologies is crucial for promoting consistency, reliability, and comparability across studies. Industry standards often align with regulatory requirements and codes, making it easier for SHM studies to meet compliance standards set by governing bodies. Then, by ensuring consistency and reliability in SHM practices, industry standards contribute to enhancing the safety of structures and infrastructure, minimizing the risk of failures or accidents. Subsequently, this scheme can enhance SHM issues like computational complexity is through promoting healthy competition and innovation among technology providers, encouraging the development of more advanced and reliable SHM systems.

4. Conclusion

This paper presents a thorough overview of the literatures concerning vibration-based Structural Health Monitoring (SHM) for steel bridges. PRISMA techniques were used to extract data from 867 citations. The research literature on SHM using vibrations for steel bridges was expanded with the review and analysis of 78 studies. It has been found that the number of publications has been steadily increasing since 2015, with the maximum record of publications directly relevant to the issues being seen in 2022 (consisting of 22 % of total studies). China (20 %) and United States of America (18 %) have found to be conducting the most explorations in these domains, followed by Australia (6 %) and Japan (6 %), Belgium (6 %). As a reliable method for keeping tabs on national infrastructures, 22 countries have contributed to studies on vibration-based structural health monitoring (SHM). A technique that is centered on vibration has been tested out on a number of different kinds of bridges. Truss-type (37 %) steel bridges were the most experiencing the applications among the many bridge types examined in the study. This research also discovered that Finite Element analysis is the most

popular option for analysing the dynamic behaviour of steel bridges by using vibration information. Numerous authors have combined the strategy with other types of analysis to broaden the scope of interdisciplinary study. Additionally, this study reveals that previous research favoured Stochastic Subspace Identification (SSI) over FDD and Enhanced FDD when implementing Operational Modal Analysis (OMA) (EFDD).

The examined literatures have mapped out the achievements of applying vibration-based technique for structural damage detection and for thorough monitoring strategies in steel bridge structures, involving implementation on a moderate level and mega projects level. From the study, shortcomings of the present studies have been outlined and discussed as a come to fore to preserve the methods' supremacy as leading structural preservation solutions. As such, the works of SHM is one of the very fast-moving realms that the future of it tend to diverse majorly from current knowledge as new technologies and research emerge beyond that point. Therefore, continuous attention should be given to revising and organizing updated knowledge in the field of structural health monitoring (SHM) because it crucially helps in determining the overall direction and progress of the SHM phenomenon.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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